TECH4BHARAT 2026 · THEME 2 JEWELTECH · PROBLEM STATEMENT 2.2 · AGENTIC AI

Saakshi

'witness', in many Indian languages

Gold never moves without two confirmations.

A neutral witness for every gram between a jeweller and a karigar: spoken receipts in their own language, fair maths, and an agent that runs every order.



Jeweller confirms + Karigar confirms = Gold moves

Team FiatLux · Shubham Jadhav · IIT Madras · Round-1 idea submission · 15 November 2026

01 · THE PROBLEM

Gold moves on trust, and disputes follow

₹40–53 lakh
of 22K gold passes through one karigar's hands every month

30–50%
of work even formal jewellery firms send out to karigars

₹2.66 crore
of gold taken in a single absconding case in Mumbai

> *"Gold issued to artisans is tracked in notebooks, leading to disputes over wastage, missed delivery dates and payments."*

PROBLEM STATEMENT 2.2

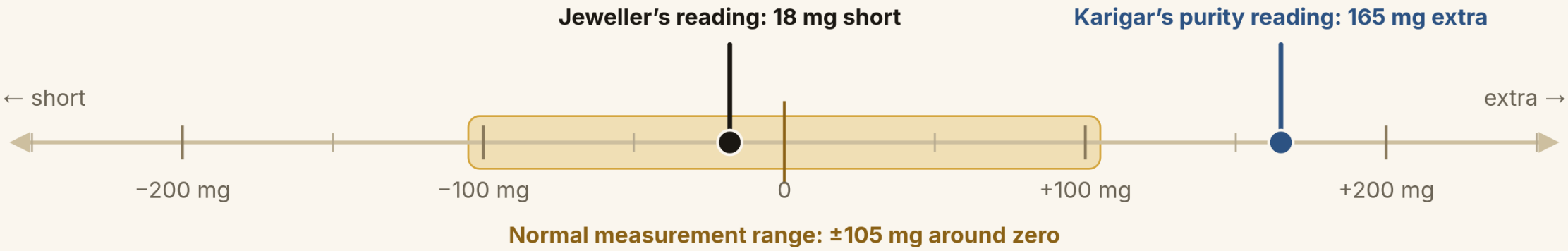
1 notebook
written and kept by the jeweller, is often the only record of all that gold

Sources: IIMA/XLRI 2023 · gold rate, Oct 2026 · press reports, Mumbai · Tech4Bharat problem statement 2.2

01 · THE PROBLEM

Honest people still fight over milligrams

In our worked example the return looks **18 mg short**. Normal measurement error is **±105 mg**.



4 parts in 1,000

A purity difference that small swings the balance by 184 mg, about ₹2,650.

Wastage is pay

It is part of the karigar’s pay, and its basis is rarely written down.

Re-tests go in circles

A similar machine has the same error, so the argument never settles.

Worked example: two 22K bangles, 48.250 g issued, 4% allowance; typical scale and purity-machine error

01 · THE PROBLEM

Today's tools serve one side only

SERVE THE JEWELLER

Paper notebook

Good at free, flexible, universal

Missing one copy, held by the jeweller; easy to alter; no reminders

Jewellery ERPs

Good at stock, billing, GST

Missing no karigar copy or confirmation; typed, English-first; every gap treated as real

Khata apps

Good at simple money reminders

Missing money only: no grams, purity or wastage

Guarantees and police checks

Good at trust inside one community

Missing slow, intrusive, local; no help with honest disputes

CONFIRMED BY BOTH



Nothing today

No record that both sides confirm, each in their own language, with fair maths for instrument error.

SERVE THE KARIGAR

Welfare IDs

Good at identity, insurance, loans

Missing prove who you are, not the work you delivered

A karigar's word stands against a notebook.

A record one side controls can't be the trail both accept.

02 · THE IDEA

Saakshi in one minute: the full flow

	Jeweller speaks Kannada	Saakshi layer by layer	Karigar speaks Marathi
1	Photographs the scale and says: “ರಫೀಕ್‌ಗೆ ಎರಡು ಬಳೆ, 22 ಕ್ಯಾರೆಟ್, ಡಿಸೆಂಬರ್ 12ರೊಳಗೆ”	→ **Channels** · the photo and voice note arrive on WhatsApp	
2	Hears it read back: 48.250 g, two bangles, 22K, for Rafiq, by 12 Dec	← **Voice and photos** · reads 48.250 g and the order, then reads them back	
3	Says “haudu” (yes): **key 1**	→ **Rules** · key 1 of 2; nothing moves yet	
4		**The agent** · plans back from 12 Dec; sends Rafiq a spoken receipt in Marathi →	“रफीकभाऊ, तुम्हाला ४८.२५० ग्रॅम सोने मिळाले आहे...”
5		**Rules** · key 2 of 2; both have said yes ←	Says “ho” (yes) or taps: **key 2**
6	Gets the receipt, code **7F3A**	← **Shared ledger** · seals one tamper-evident entry for both →	Gets the same receipt, code **7F3A**

THEN THE AGENT checks in → re-plans if work slips → checks the return in pure gold → both approve the settlement → UPI payment

02 · THE IDEA

An agent runs every order, and asks first

ACROSS THE LIFE OF EACH ORDER

Plan
Works back from the due date and picks the right karigar.

→

Act
Sends receipts, voice check-ins, statements and UPI links.

→

Recover
No reply? A reminder, a call, a café visit. Late? A new plan.

→

Reconcile
Checks the return in pure gold and explains it to both.

→

Settle
Pays per job, not at month end, and tracks the wait.

Remember Under every step: each order's full history, and each person's language, terms and preferences.

Always asks first

Both sides confirm every issue, return, change of terms, settlement and dispute. The jeweller approves customer updates. Only the karigar can share their record.

02 · THE IDEA

Every language, every phone, any AI

All 22 scheduled languages + English

Each side hears the same record in their own language.

CHECKED BY NATIVE SPEAKERS AT LAUNCH

- **Bengali**
- **English**
- **Hindi**
- **Kannada**
- **Marathi**
- **Tamil**

CHECKED AS PILOTS GROW

Assamese · Bodo · Dogri · Gujarati · Kashmiri · Konkani · Maithili · Malayalam · Manipuri · Nepali · Odia · Punjabi · Sanskrit · Santali · Sindhi · Telugu · Urdu

Speak and listen

Voice first, buttons as backup. No typing, and no new app for the karigar.

Entry-level phones

Works on basic Android and patchy networks. The jeweller's app works offline.

Assisted mode

JanAI Café operators and Saakshi points can onboard and confirm, with consent.

Any AI model

Cloud, India-built, or fully offline on the shop's own computer. Gold and money rules never depend on the AI.

03 · HOW IT WORKS

How Saakshi fits together

1	**Channels**	WhatsApp voice and buttons, calls, a simple app, café desks
2	**Voice and photos**	22 languages heard and spoken; scale photos read; everything read back
3	**The agent**	Plans, follows up, re-plans, reconciles and settles; remembers every case
4	**Rules and approvals**	Decides who must confirm what. Only rules and both parties' confirmations write to the ledger.
5	**Shared ledger**	Tamper-evident and identical for both; corrections are new entries

AROUND EVERY LAYER

Consent vault

Consent by voice, per purpose, in each person's language. Minimal data.

Any AI model

Cloud, India-built or fully local, swapped without changing the product.

Partners

UPI payments, hallmarking centres, later lenders and insurers, with consent.

03 · HOW IT WORKS

Fair by design

Returns are checked in pure gold, allowing for normal instrument error.

Matches the agreement

Settle as agreed.

Within normal measurement range

Nobody is accused. The rule both sides agreed in advance applies.

Needs a re-check

Neutral re-weigh or re-test at a Saakshi point, with both present.

Wastage is pay

Terms are co-signed and never tightened silently.

Flag the scale

The same gap for every karigar flags the scale, not the people.

No blacklists

Disputes are settled by both sides, with shared evidence.

“Mera hisab”

Any karigar can hear their own account at any time.

Karigar passport

A clean record the karigar owns and shares only by consent.

04 · BUSINESS AND PLAN

Jewellers pay. Karigars never do.

FREE
₹0
up to 3 karigars

PRO
₹999
a month · up to 15 karigars

BUSINESS
₹1,999
a month · unlimited karigars, staff

MANUFACTURER
₹100–150
a karigar a month · large networks

Karigars are always free. Prices are hypotheses we are testing in the field.

FOR ₹999 A MONTH

≈ ₹66,000

saved a month by cutting unexplained variance 0.25% on 2 kg issued

2.37 lakh — BIS-registered jewellers across 392 districts

₹4.54 lakh crore — India’s gold jewellery demand in 2025

Later revenue — Credit and insurance on consented records, a wedding-season marketplace, loss benchmarks

Sources: BIS (PIB, Nov 2026) · World Gold Council 2025 · ROI illustrative, to be checked in pilots

04 · BUSINESS AND PLAN

Grows cluster by cluster, creating jobs

1. **Start in Bengaluru**
   Avenue Road, Chickpet, Nagarthpet: Kannada, Bengali and Hindi speakers side by side.

2. **Enter through trusted bodies**
   Jewellers' and karigar associations; testing shops and hallmarking centres become Saakshi points.

3. **Let karigars invite**
   Every karigar works for several jewellers and can invite each of them.

4. **Shadow pilots, no risk**
   Two weeks alongside the notebook, then a value report: grams confirmed, on-time %, days to pay.

5. **Next clusters**
   Chennai, Coimbatore, Thrissur, Mumbai, Kolkata.

Jobs it creates

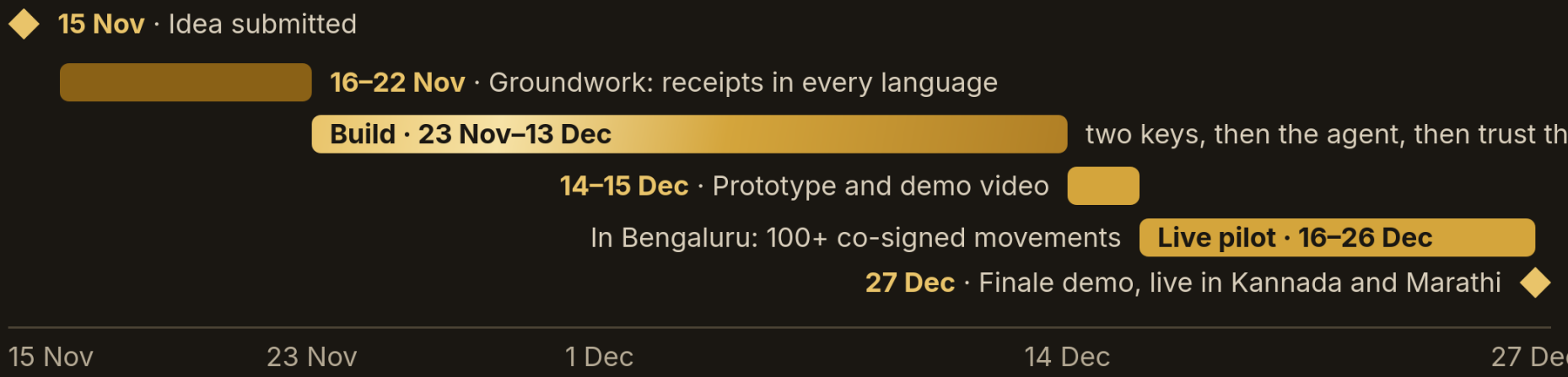
Saakshi Mitras
Local youth who onboard and support jewellers, about one per 100, paid per active jeweller.

Café operators
JanAI Café and Saakshi-point operators earn per assisted service.

Karigars who grow
Faster pay, direct work and credit to open workshops and hire apprentices.

04 · BUSINESS AND PLAN

From idea to a live pilot by 27 December



What we will prove at the finale

- ✓ Every gold movement confirmed by both sides
- ✓ No accusation over normal measurement error in our test cases
- ✓ One order end to end, live, in two languages
- ✓ The same receipt in all 23 languages

TEAM

FiatLux

Shubham Jadhav, **IIT Madras**

founder: product, field research, design and build. Native speakers review each launch language.